Data in Brief 54 (2024) 110524



Contents lists available at ScienceDirect

Data in Brief

journal homepage: www.elsevier.com/locate/dib



Data Article

# Dataset of chilli and onion plant leaf images for classification and detection



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ARTICLE INFO

*Article history:*
Received 10 February 2024
Revised 19 April 2024
Accepted 7 May 2024
Available online 15 May 2024

Dataset link: Dataset of Chilli and Onion Plant Leaf Images for Classification and Detection (Original data)

*Keywords:*
Classification of leaf diseases
Image dataset
Diagnosis of disease

ABSTRACT

This article presents the chili and onion leaf (COLD) dataset, which focuses on the leaves of chili and onion plants, scientifically known as Allium cepa and capsicum. The presence of various diseases such as Purple blotch, Stemphylium leaf blight, Colletotrichum leaf blight, and Iris yellow spot virus in onions, as well as Cercospora leaf spot, powdery mildew, Murda complex syndrome, and nutrition deficiency in chili, have had a significant negative effect on onion and chili production. As a consequence, farmers have incurred financial losses. Computer vision and image-processing algorithms have been widely used in recent years for a range of applications, such as diagnosing and categorizing plant leaf diseases. In this paper we introduced a detailed chilli and onion leaf dataset gathered from Chilwadigi village with varying climatic conditions in Karnataka. The dataset contains a variety of chili and onion leaf categories carefully selected to tackle the complex challenges of categorizing leaf images taken in natural environments. Dealing with challenges such as subtle inter-class similarities, changes in lighting, and differences in background conditions like different foliage arrangements and varying light levels. We carefully documented chilli and onion leaves from various angles using high resolution camera to create a diverse and reliable dataset. The dataset on chilli leaves is set to be a valuable resource for enhancing computer vision algorithms, from traditional deep learning models to cutting-edge vision transformer architectures. This will help in creating advanced image recognition systems specifically designed for identifying chilli plants. By mak-

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https://doi.org/10.1016/j.dib.2024.110524
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ing this dataset publicly accessible, our goal is to empower researchers to develop new computer vision techniques to tackle the unique challenges of chilli and onion leaf recognition. You can access the dataset for free at the following DOI number: http://doi.org/10.17632/7nxxn4gj5s.3 and http://doi.org/10.17632/tf9dtfz9m6.3

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Specifications Table

| | |
|---|---|
| Subject | Two fields: Agricultural science and computer science. |
| Specific subject area | Image processing |
| Type of data | Raw data, images |
| How the data were acquired | The data was collected using an APS-C camera with a Canon Mark II D lens with measurements of 122.4 × 92.6 × 69.8 mm. It has a maximum continuous shooting rate of 5 frames per second, a resolution of 72 dots per inch horizontally and vertically, a bit depth of 24, a monitor of 3.0 inches, and autofocus. A pathologist assisted with data gathering, and data interpretation was done manually. |
| Data format | JPG Format |
| Description of data collection | Photographs are gathered during both rainy and sunny weather conditions. A dataset was developed consisting of 864 raw photos of onions and 2064 images of chili plant leaves. |
| Data source location | • Data collection is conducted in the Chilwadigi village onion farm field located in the Koppal district of Karnataka. The geographical coordinates of this field are Latitude: 15.39783 and Longitude: 76.161604.<br>• The Chilwadigi village is located in the Koppal area. It is known for its chilli farm field, which is situated at a Latitude of 15.618803 and a Longitude of 76.027645. |
| Data accessibility | Repository name: Mendeley Data<br>Data identification number of Chilli leaf dataset: http://doi.org/10.17632/tf9dtfz9m6.3<br>Data identification number of Onion leaf dataset: http://doi.org/10.17632/7nxxn4gj5s.3<br>Direct URL to data: https://data.mendeley.com/datasets/tf9dtfz9m6/2 |

## 1. Value of the Data

- The dataset enables the researcher to develop computer-vision algorithms that exhibit exceptional performance in real-time scenarios, as the dataset was acquired from authentic farming environments.
- The dataset can be utilized in Machine Learning or Deep Learning algorithms to enhance various tasks such as disease classification, segmentation of diseased patches on leaves, evaluation of disease severity, and estimation of leaf area, among others.
- Researchers will find the dataset to be quite challenging, as it contains images taken in real-time situations, but it also contains enough information to build plant leaf disease categorization applications that farmers can use to help them crops more effectively.

## 2. Introduction

Computer vision, a technology focused on analyzing and comprehending real-time visual information, emulates this capability, although to a limited degree. Computer systems can achieve

higher intelligence in computation and effectively tackle complex problems by emulating human vision and processing real-world information found through cameras. The fundamental steps of digital image processing, which include image collection, processing, feature extraction and object identification, facilitate the understanding of visual data. Computer vision has many uses in many different industries, such as smart agriculture and medical diagnostics. The third green revolution, or smart agriculture, is a way of farming that uses cutting-edge technology to boost yields and the quality of crops. This includes things like drones, big data, sensors, cameras, and the Internet of Things (IoT). Fields host devices like sensors and cameras to collect data.

In this context, digital images captured by cameras are critical. Plant photographs can capture a range of elements, such as stages of growth, illnesses, life cycles, soil conditions, and vegetation indices. Plants are crucial for sustaining a balanced ecosystem as they provide oxygen, materials for construction, and nourishment for various organisms. Similar to any other living organism, plants can become ill due to factors such as fungal infections, bacterial infestations, and viral attacks, extreme variations in temperature and humidity, and inadequate nutrient supply. Diseases and disorders that have a negative impact on plant health can significantly diminish economic and societal value. Researchers have explored various approaches involving computer vision techniques, the processing of digital images, and soft computing strategies.

This paper offers a data collection of chili and onion leaves, diagnosis of chili and onion leaf diseases, and approaches used in diagnosing symptoms of diseases through the analysis of plant leaf images. This article presents several investigations on the recognition and classification of diseases in plant leaves, aiming to encourage further research in this field. Our primary objective is to offer researchers to work on detecting plant diseases. Another objective is to encourage environmental sustainability through efficient plant development management. This study has numerous implications:

- Utilizing machine learning models trained on a chili and onion plant leaf dataset has the potential to revolutionize disease detection in chili crops. By automating the process, farmers can identify diseases early on, enabling prompt intervention to minimize crop losses.
- A dataset of chili and onion plant leaves is a valuable resource for researchers, agronomists, and policymakers interested in studying the spread and patterns of plant diseases in chili and onion crops. Through analyzing patterns and trends in disease occurrence and severity, stakeholders can develop specific strategies for disease management and crop protection.

Furthermore, having a standardized dataset of chili and onion plant leaves makes it easier to conduct benchmarking studies and compare various disease detection algorithms and methodologies. This allows researchers to assess their models' performance objectively and pinpoint areas for enhancement, encouraging creativity and teamwork in the agricultural technology sector. Given the above implications, it is evident that this study's significance lies in creating a chili plant leaf dataset to tackle the challenges presented by plant diseases in chili cultivation. Utilizing technology and data-driven methods, we can equip farmers with the necessary tools and information to protect chili crops and guarantee food security for the coming generations.

## 3. Comparative Analysis

When comparing two different image datasets, namely the plant village (PV) dataset, chilli leaves dataset created by Mahaning Hubballi from Kaggle, with a well-established dataset, and a newly collected dataset of chilli and onion leaves images called COLD, the differences and practical implications for image classification are significant. The Plant Village dataset comprises 2475 small chili datasets with standardized image dimensions of $256 \times 256$, categorized into 2 classes. Additionally, there is another chili dataset from Kaggle, consisting of 6166 files with image dimensions of $500 \times 333$, categorized into 5 classes. However, the creator of this dataset did not label the data. This dataset offers a controlled environment that is well-suited for evaluating and comparing image classification algorithms. On the other hand, the COLD dataset has 2928

**Table 1**
The state of the onion leaf, together with the total number of each state.

| Category | No of Images |
|---|---|
| Healthy | 1278 |
| Iris yellow virus | 1272 |
| Purple Blotch | 735 |
| Leaf Blight | 1217 |
| Total | **4502** |

**Table 2**
The state of the chilli leaf, together with the total number of each state.

| Category | No of Images |
|---|---|
| Healthy | 2198 |
| Cercospora | 2219 |
| Mites and trips | 2507 |
| Nutritional Deficiency | 2032 |
| Powdery Mildew | 4502 |
| Total | **10,987** |

different photographs taken from camera without augmentation and 10,987 images with augmentation as shown in Tables 1 and 2, which provide a diversified collection of more intricate and authentic scenarios that encompass a broad spectrum of infected images in natural environments. The existing dataset is suitable for basic machine learning projects and instructional use since it is simple and consistent. However, the COLD dataset is more appropriate for applications that require strong performance in real-world situations due to its authentic imagery and the dataset's source in natural settings makes it highly relevant and applicable for addressing real-time challenges in agricultural production and crop monitoring. Hence, the choice of these datasets should be determined by the particular requirements of the research; The current dataset is suitable for evaluating the effectiveness of algorithms in controlled environments. On the other hand, the COLD dataset is more suitable for developing technologies that are intended for real-world use, particularly in fields that require high accuracy in various and unpredictable conditions. The Xception model, trained on the COLD dataset, achieves an accuracy of 79.56 % and 88.55 % accuracy without using any augmentation technique. This indicates superior generalization to real-world scenarios, as the dataset contains different and complicated images. The COLD dataset is expected to exhibit greater diversity in terms of diseases, lighting conditions, and environmental factors compared to the Existing dataset.

## 4. Background

The study of plant disease detection amalgamates plant pathology, agronomy, remote sensing, image processing, and machine learning. Traditional visual inspection methods are being supplanted by technological advancements. Remote sensing tools, including satellites and UAVs, offer holistic views of crop health. High-resolution imagery from drones enhances disease monitoring in real-time. Image processing algorithms extract valuable information from raw data. Machine learning, particularly CNNs, aids in automated disease diagnosis. Challenges persist, including data scarcity and algorithm generalization. Interdisciplinary collaboration is crucial for progress. Solutions aim to enable early disease detection and effective management. Ultimately, the goal is to safeguard global food security and sustainable agriculture.

## 5. Data Description

All the photographs have been gathered from Chilwadigi village, located in the Koppal district. Data is collected during the Kharif season. The photographs are acquired utilizing a

Canon Mark II D digital camera in rural landscapes. The chili leaf photos are classified into five main categories based on their state. The collected images undergo pre-processing and are then stored in five separate folders based on their category. The "healthy" folder contains 2198 images, the "Cercospora" folder contains 2219 images, the "Mites and trips" folder contains 2507 images, the "Nutritional Deficiency" folder contains 2032 images, and the "Powdery Mildew" folder contains 2031 images. There are a total of 10,987 photos stored in the cropped and resized folder. The dataset of onion leaf photos is organized into four folders: "healthy" has 1278 images, "Iris yellow virus" contains 1272 images, "Purple blotch" contains 735 images, and "Leaf Blight" contains 1217 images. Plant pathologists assist in labeling the images, and the dataset can be accessed publicly at the following links: https://data.mendeley.com/preview/tf9dtfz9m6?a = 91201c66-9430-4d12-9469-1288e5e6b858 and https://data.mendeley.com/preview/7nxxn4gj5s?a = 913f5f4d-da0d-44e6-8521-8fa235913b02

### 5.1. Details of onion plant leaves

All the pictures from the Koppal area, including the image of the Kharif season, were collected. Five different types of onion plant leave include Purple blotch, Stemphylium leaf blight, Colletotrichum leaf blight, and Iris yellow spot virus. Below are descriptions of various onion leaf diseases:

#### 5.1.1. Purple blotch

Purple blotch disease, caused by *Alternaria porri*, is a significant issue in agriculture due to its severe damage to Allium spp. crops and substantial reduction in production. Symptoms manifest 1–4 days post-infection, starting with bulb rot those progresses into dark reddish-purple and subsequently brownish-black lesions. Various elements such as season, sowing time, humidity, temperature, crop stage, and plant design significantly influence the development of purple blotch disease. Fig 1 illustrates the comprehensive life cycle of purple blotch disease.

#### 5.1.2. Stemphylium leaf blight

Stemphylium leaf blight is a fungal disease that affects onions and is caused by the fungus Stemphylium vesicarium. This virus flourishes in surroundings that are both warm and humid, and its spores can be disseminated through the action of wind, rain, or contact with surfaces that have been polluted. The disease presents itself as distinctive lesions on the leaves of onions, initially appearing as small, water-soaked patches that subsequently grow in size and acquire a tan to grayish coloration, accompanied by a dark border. These lesions have the ability to combine, resulting in significant harm. Stemphylium leaf blight has a detrimental impact on onion foliage, resulting in the yellowing, withering, and eventual mortality of the leaves. Severe infections can lead to substantial reductions in crop output, hence affecting the overall quality and productivity of onion crops. As the disease advances, it hampers the process of photosynthesis, diminishing the plant's capacity to produce energy and hindering its growth as depicted in Fig 2.

#### 5.1.3. Colletotrichum leaf blight

Anthracnose, a disease caused by Colletotrichum, is easily identifiable by its symptoms that appear on onion leaves. Tiny, deep sores bordered by a dark line appear on infected leaves. These lesions have a tan, dark brown, or black center that can enlarge and harm tissues over time. Image of disease shown in Fig 3.

#### 5.1.4. Iris yellow spot virus

Iris Yellow Spot Virus (IYSV) is a pathogenic viral disease that affects onions. The disease has the potential to cause severe damage to an onion field by impairing the photosynthetic area of the leaves of infected plants (Fig. 4). This, in turn, limits the plant's capacity to grow larger bulbs, thus impacting the overall yield and quality of onions upon harvest.

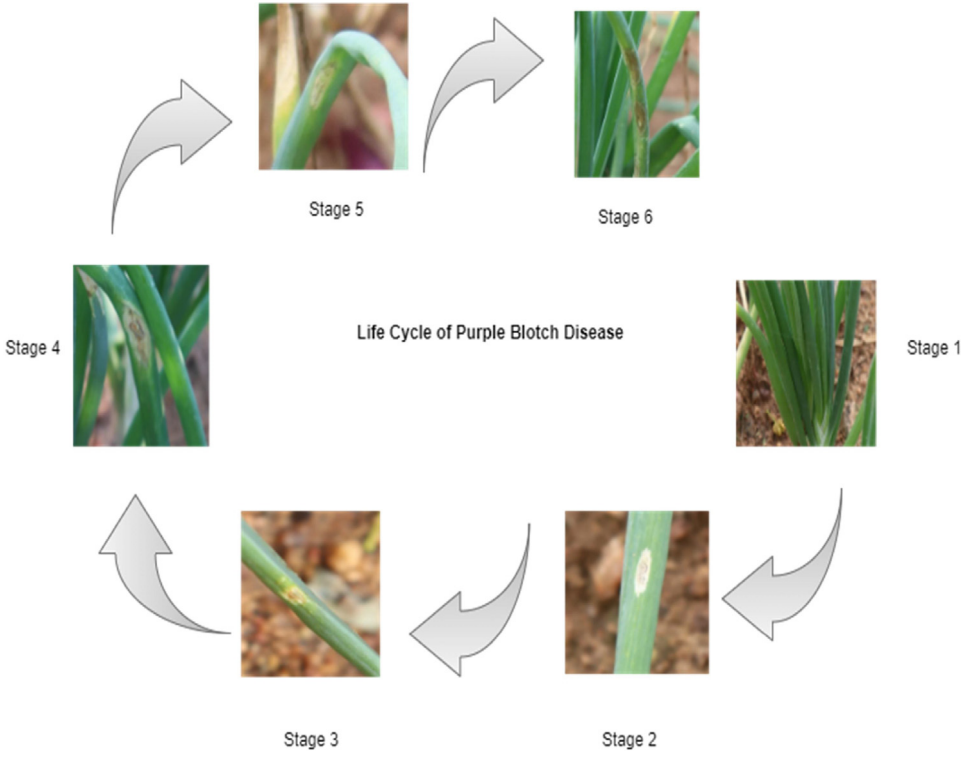



**Fig. 1.** Life cycle of Purple Blotch Disease.



**Fig. 2.** Stemphylium leaf blight.

### *5.2. Details of chilli plant leaves*

All of the images from the koppal region, during the kharif season image are gathered. There are four distinct categories in chili plant leaves they are Cercospora leaf spot, powdery mildew,



**Fig. 3.** Colletotrichum Leaf Blight.



**Fig. 4.** Iris Yellow Spot Virus.

Murda complex syndrome, Nutritional deficiency and healthy leaves. The description of individual chili leaf diseases are given below:

*5.2.1. Cercospora leaf spot*

Cercospora mostly spreads by infected seeds, although it can also persist in plant residue and soil for a minimum of one year. Spores are mostly dispersed through wind, splashing water, and contact between leaves. The optimal climatic conditions for the abundant growth of Cercospora leaf spot are a temperature below 28°C, a relative humidity of 92%, and a soil pH ranging from 5 to 6. The disease does not occur when the relative humidity is below 90%. The severity of the disease is greater during periods of rainy weather compared to dry weather, and it becomes highly damaging when the relative humidity is high [1]. The presence of Cercospora leaf spot disease disrupts the photosynthetic process and causes deformation of the leaves. This weakens the plant and leads to premature defoliation, ultimately reducing the production and market value of the plant [2]. Typically, signs of Cercospora leaf spot (CLS) disease are visible on both surfaces of the leaf. The early phase of the infection is characterized by the presence of small, circular spots about 0.5–1 mm in diameter. These spots have a dark brown color, with a light-grey center and reddish-brown edges, resembling an ashy appearance. These symptoms primarily appear on the leaves. As the condition advances, the spots increase in size, reaching a diameter of up to 1.5 cm. Dark concentric rings form around the whitish center, creating the



**Fig. 5.** Cercospora leaf spot.



**Fig. 6.** Powdery mildew.

distinctive 'frog-eye' symptom. Over time, the spots come together and the center, which is a pale gray color, often becomes dry and falls out, resulting in a shot-hole look. The leaves undergo a transformation, changing from green to yellow, withering, and eventually falling off, so leaving the fruit vulnerable to sunscald [3]. Image of Cercospora leaf spot is displayed in Fig 5.

*5.2.2. Powdery mildew*

Powdery mildew is a fungal infection that impacts chili plants, causing the development of a white, powdery substance on the leaves and other parts of the plant. The responsible organisms frequently belong to the order Erysiphales. The disease flourishes in hot and arid environments and is transmitted through airborne spores. Chilli plants suffer from diminished photosynthesis and leaf deformation due to the detrimental effects of powdery mildew. Management measures encompass the implementation of adequate air circulation, application of fungicides, and the careful selection of chili types that possess resistance. Prompt identification and timely action are crucial for efficient management of powdery mildew in chili plants. Image of Powdery mildew is displayed in Fig 6.



**Fig. 7.** Murda Complex Syndrome.



**Fig. 8.** Nutritional deficiency.

*5.2.3. Murda complex syndrome*

Chilli leaf curl, also known as murda, is an extremely damaging disease that affects chilli plants in India. Murda in chilli is a highly intricate ailment triggered by pathogens such as thrips, mites, and viruses [4]. Chilli murda is a multifaceted condition linked to the presence of thrips, mites, and viruses. The curling of leaves is a symptom of the murda disease, which is commonly known as chilli leaf curl due to this particular symptom and image of Murda complex syndrome is displayed in Fig 7.

*5.2.4. Nutritional deficiency*

The most noticeable symptom is interveinal chlorosis, which is characterized by the gradual yellowing of older leaves, beginning at the margins and moving inward while the veins remain green. As the yellowing progresses, the leaves may begin to curl and cup, and eventually fall off the tree before their time. The plant's growth is hindered by the deficit, leading to stunted development. Image of Nutrition deficiency is displayed below in Fig 8.

## 6. Experimental Design, Methods and Materials

### 6.1. Data collection

This section provides a comprehensive explanation of the methodical methodology used to acquire the chili and onion leaf dataset in Chilwadigi village, Koppal district, Karnataka. Using a strategic approach, we publish our dataset, which includes photos of chili and onion leaves. This dataset is now easily accessible through the Mendeley data repository at the following link: https://data.mendeley.com/preview/tf9dtfz9m6?a=91201c66-9430-4d12-9469-1288e5e6b858 and https://data.mendeley.com/preview/7nxxn4gj5s?a=913f5f4d-da0d-44e6-8521-8fa235913b02.

### 6.2. Data acquisition

This section offers a detailed explanation of the thorough examination and careful selection method used for the chili and onion leaf images. We employed a rigorous approach to carefully examine and remove photos that displayed blurriness, noise, or lack of focus. This process guaranteed that only photographs with excellent resolution were included in the dataset. In addition, we created a Python script that utilizes the scikit-image library package to ensure uniformity and standardization among image samples. This script enabled the adjustment of the size of all chosen photos to a specific dimension of 224 $\times$ 224 pixels. The carefully selected photographs in our dataset have been thoroughly processed to ensure the highest quality and consistency. They can now be accessed through Mendeley Data [7].

### 6.3. Data pre-processing

#### 6.3.1. Cropping and resizing

Individuals involved in the field of Deep Learning can enhance their work by utilizing a cropping tool to focus on the specific area of interest within a leaf image. Deep Learning models typically require input images to have a consistent size. Therefore after cropping, the images are reduced to dimensions of 224$\times$224. Fig 9 depicts the cropped images of onion and chilli leaf.

#### 6.3.2. Data augmentation

Image data has been increased using augmentation technique such as rotating, flipping, zooming and shearing to artificially increase the diversity of the dataset, thereby improving the robustness, generalization, and applicability of models and algorithms trained on these datasets [8]. Fig 10 display the augmented images.

#### 6.3.3. Importance of the dataset

A comprehensive dataset on chilli and onion crops holds immense significance for agricultural optimization and sustainability. By facilitating the early detection and management of diseases, this dataset enables farmers to apply precise interventions, reducing crop damage and improving yield quality. Furthermore, the insights derived from analyzing plant health data help in optimizing agricultural practices, ensuring efficient resource use. Such datasets not only support individual farmers in enhancing their productivity but also contribute to broader food security goals by enabling more resilient and robust agricultural system.

The primary rationale for our study was to address the challenges associated with training deep learning models on small datasets, a common obstacle in fields where large-scale data collection is impractical or impossible so we employed the transfer learning approach to train our small dataset by utilizing the Xception model, which had already been pre-trained on the ImageNet dataset. Leveraging a pre-trained model allowed us to capitalize on the learned features

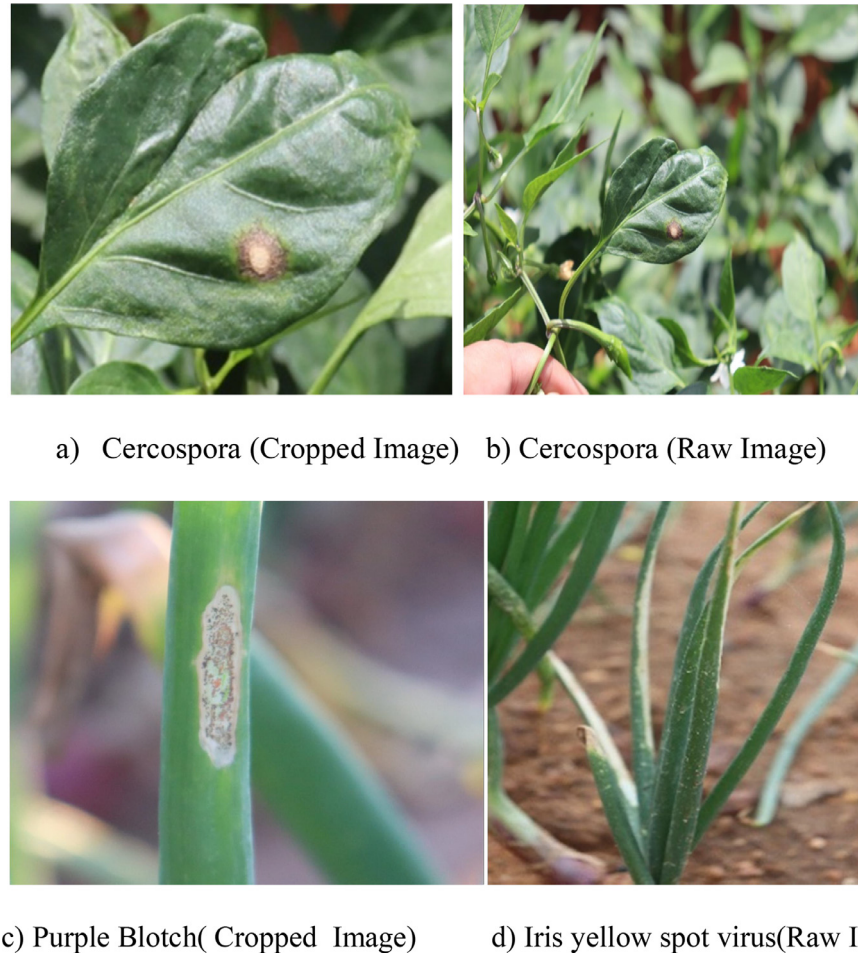



**Fig. 9.** Examples of cropping and resizing of chilli and onion leaves.

from a vast and diverse set of ImageNet images, which can effectively generalize across a wide range of visual tasks. This pre-training serves as a foundational knowledge base for the model, enhancing its ability to adapt to our specific task with a relatively smaller dataset. Importantly, the pre-training on ImageNet did not negatively influence our training process. Instead, it provided a significant head start by not requiring the model to learn features from scratch, thus speeding up the learning process and improving the overall performance on our dataset

## 7. Specifics of Implementation

### *7.1. Xception*

The Xception framework, proposed by F. Chollet, is influenced by the Inception 3.0 architecture [5]. Both architectures possess an equal amount of model parameters, which is worth mentioning. However, the Xception architecture optimizes the utilization of these factors. The authors have shown that the discrete spectrum includes depth-wise separate convolutions and



a) Cercospora (original) b. Cercospora (Augmented)



c) Iris Yellow Spot Virus(Original) d) Iris Yellow Spot Virus (Augmented)

**Fig. 10.** Images of onion and chilli leaves using augmentation techniques.

point-wise convolutions at both ends, while the inception modules are positioned in between. The inception modules were replaced with depth-wise separable convolutions, leading to enhanced classification performance without any additional processing burden. The basic structure of the Xception model is illustrated in Fig 11.

### *7.2. Result analysis*

The COLD dataset [6,7] is utilized to train and evaluate the performance of the Xception model in disease identification and categorization. The chili dataset comprises 532 raw images without augmentation that have been cropped and resized with five classes. On the other hand,

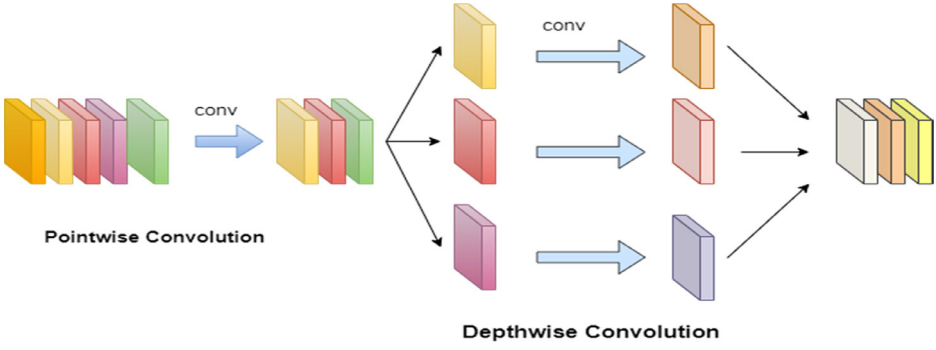



**Fig. 11.** Architecture of Xception.

**Table 3**
Classification chart of Chilli leaf dataset.

| | Precision | Recall | F1-Score |
|---|---|---|---|
| Cercospora leaf spot | 0.9595 | 0.7100 | 0.8161 |
| Healthy | 0.6154 | 0.6400 | 0.6275 |
| Mites and thrips | 0.5556 | 0.9375 | 0.6977 |
| Nutritional deficiency | 0.7895 | 0.8571 | 0.8219 |
| Powdery mildew | 0.9714 | 0.9714 | 0.9714 |
| Accuracy | | | **79.56%** |

**Table 4**
Classification chart of Onion leaf dataset.

| | Precision | Recall | F1-Score |
|---|---|---|---|
| Healthy | 0.8330 | 0.8849 | 0.8582 |
| Purple Blotch | 0.9831 | 1.0000 | 0.9915 |
| Iris Yellow Spot Virus | 0.8717 | 0.8051 | 0.8371 |
| Accuracy | | | **88.55%** |

**Table 5**
Associated studies using preexisting datasets on onions and chilli.

| Sl.No | Objective | Dataset | Remarks |
|---|---|---|---|
| 1 | Detection and Classification of Chilli Leaf Diseases [9] | 5100 images | Locally collected |
| 2 | Chilli leaf Dataset [10] | 6166 images | Locally collected |
| 3 | Image based Onion Leaf Disease detection using deep convolution neural network [11] | 1000 images | Locally collected |

the onion dataset has 10,149 images including augmented and raw images that are labelled with three classes. The Xception model, with RMSprop acting as optimizer, has accomplished overall validation accuracy on chili and onion dataset is 79.56 % and 88.55 %. The classification outcome is presented in Tables 3 and 4.

By analyzing Fig. 12, the confusion matrix of the chili dataset indicates that the model exhibits some ambiguity in distinguishing between a leaf affected by mites and a leaf affected by Cercospora spot disease. While a few photographs depicting powdery mildew were mistakenly classified as Cercospora spot disease, which led to an accuracy rate of 79.56 %.

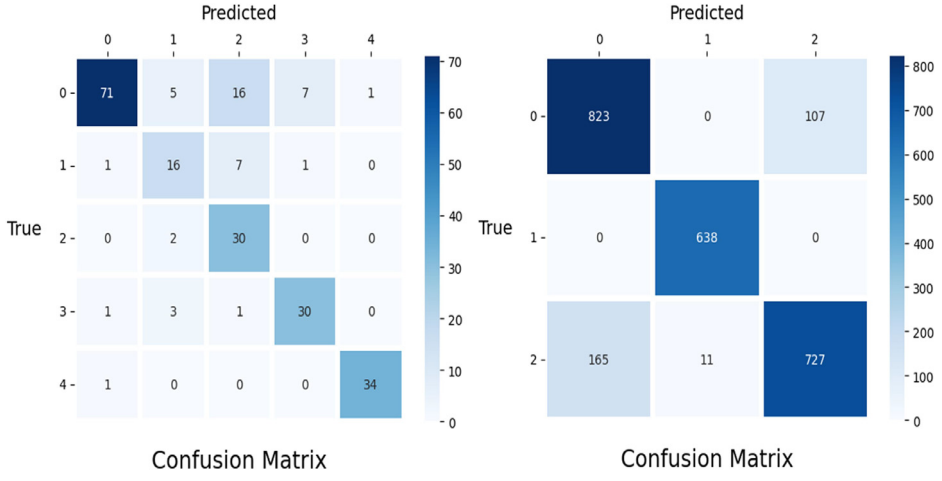



**Fig. 12.** Confusion matrix of Xception classifier using chilli dataset and onion dataset.

The confusion matrix of the Xception classifier using the onion leaf dataset indicates that the model successfully identified purple blotch but misclassified it between healthy and Iris yellow spot viruses.

## 8. Conclusion

In our study of chili and onion plant diseases, we identify crucial gaps in understanding and highlight opportunities for future research. We emphasize the potential for practical solutions to benefit the chili and onion farming industry, leveraging advancements in technology and promoting sustainable practices. Collaboration between researchers, industry stakeholders, and policymakers is key to addressing these challenges effectively. Our data article presents a new chili and onion dataset featuring high-resolution images showcasing various diseases (Table 5). This dataset differs from existing chili and onion leaf datasets and provides valuable insights for improving knowledge and practices in managing chili plant diseases, with significant implications for industry and sustainability.

## Limitations

- Different stages and kinds of onion and chili plants exhibit distinct symptoms, making disease detection more complicated.
- Variables such as humidity and temperature have an impact on the appearance of diseases, making it more challenging to diagnose them.

## Ethics Statements

1. There is no other publication of this article; it is completely unique.
2. The article is currently not being considered for publication elsewhere.
3. This article is an accurate and comprehensive reflection of the writers' personal research and analysis.

We agree with the above statements and declare that this submission follows the policies of Solid State Ionics as outlined in the Guide for Authors and in the Ethical Statement.

## CRediT Author Statement

**Aishwarya M P:** Data Collection and Writing, **Dr. A. Padmanabha Reddy:** Reviewing and Editing.

## Data Availability

Dataset of Chilli and Onion Plant Leaf Images for Classification and Detection (Original data) (Mendeley Data).

## Acknowledgment

Authors would like to thank Dr. M R Govindappa, College of Agriculture, Hagari, Karnataka, India.

## Declaration of Competing Interest

The authors affirm that they do not possess any identifiable conflicting financial interests or personal ties that could have potentially influenced the findings presented in this paper.

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